

Vartan Chakryan

818-509-4404 | vartanchakryan@gmail.com | <https://www.linkedin.com/in/vartan-c1/>

U.S. Citizen | Los Angeles, CA

Education & Certificates

- M.S. Computer Science, Georgia Institute of Technology - Remote | In Progress
- B.Sc. Computer Engineering, Cal State University Northridge | Graduated

Technical Skills

- **Languages:** C, C++, Python (pandas, NumPy)
- **Systems Programming:** Bare metal firmware, RTOS, register level peripheral configuration
- **Embedded Protocols:** GPIO, PWM, Timers, UART, I2C, TCP/IP, UDP
- **Microcontrollers & Hardware:** ARM Cortex M7, ARM Cortex M4, electrical schematic reading, soldering, breadboarding
- **Toolchain & Version Control:** Git, GitHub, Docker, Wireshark, VS Code, Linux, CMake
- **Databases:** PostgreSQL

Experience

Hardware Lead, Autonomous Drone, CSUN SAE Aerospace Design Team

- Led a cross functional team building a fully autonomous UAV that used computer vision and GPS to locate a target and deploy a payload with precision.
- Directed hardware integration across flight controllers, GPS, ESCs, and telemetry systems, ensuring reliable real time communication between subsystems under flight conditions.
- Coordinated task allocation and mentored engineers across avionics, software, and mechanical subteams to deliver a competition ready system on a fixed deadline, developing communication and leadership skills directly applicable to cross disciplinary engineering teams.

Project: Autonomous Two Wheel Embedded Robot

- Designed and implemented a real time embedded system in bare metal C on an ARM Cortex M4, using zero hardware abstraction libraries, configuring GPIO, timers, PWM, and UART directly at the register level.
- Built an ultrasonic based distance sensing and motor control loop, demonstrating hands on systems and low level programming fundamentals relevant to flight software development.
- Implemented live UART telemetry output for real time debugging and validation, mirroring the test and verify workflow used in flight software integration.